One Person Company (OPC) AI Agents system with multi-modal fusion for health assistance

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***Abstract*—This paper designs and implements a multi-agent system based on the One Person Company (OPC) AI architectural concept. This system leverages large language models (LLMs) and multiple agents, combining multi-modal information, with a new AI agent orchestration framework to coordinate multiple AI agents. The application is for high-altitude stroke management. Using our system, a single human physician directs six specialized agents to perform health assistance tasks. The system operates pre-screening on the risk of stroke, dynamic physiological monitoring, FAST screening, and data fusion. By processing various types of data, such as text, physiological signals, and audio-visual information, our OPC AI system achieves intelligent risk assessment while maintaining a human-led and AI-agent-executed workflow.**

***Index Terms*—One Person Company (OPC) AI, Multi-agent System, Health Assistance, Multi-modal Data Fusion**

I. Introduction

A new entrepreneurial concept, "OPC AI Agent System", refers to a One Person Company (OPC) [1]. It is an emerging new AI design: it operates primarily through the use of Artificial Intelligence Agents (AI Agents): a person acts as the "strategic brain" to decide the functionality of an AI system [2]. This person allocates tasks to a team of AI Agents that function as "digital employees", each agent having a specific skill set or expertise. This person also manages these agents to coordinate teamwork. Application examples include those in computing, data, law, finance, and other fields [3].

Most physicians are manually working on medical cases, demanding AI assistants to improve efficiency [4]. For example, build OPC AI agents that act as research assistants for physicians, helping them with tasks such as natural language processing, organizing expert knowledge, handling data management, and intelligent Q&A.

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The proposed AI agent system using the OPC AI concept should exploit artificial intelligence to interact with human users. such as the doctors, medical researchers, and nurses. The system will take various inputs, including text such as users' questions, prompts, messages, voice such as audio from a patient, videos such as one for a patient's walking posture, as shown in Fig.1, and design multi-modal fusion methods [5] to provide sophisticated decision making.

Our OPC AI system has demonstrated versatile capabilities, which understands users and considers various inputs in different formats. This research offers a promising potential for future AI assistance in health care.

II. Related work

A. Multi-agent system

Recent research on multi-agent systems (MAS) emphasizes sophisticated coordination mechanisms enabling agents with diverse tasks to collaborate effectively. A key development is the rise of Large Language Model (LLM)-based agents, which introduce new paradigms for communication and task distribution [6] [7]. A communication-centric view is now central to understanding coordination. Research considers system-level communication (structures and protocols) and system internal communication (strategies and content) that allows agents to negotiate and achieve collective intelligence [8]. Other taxonomies categorize collaboration mechanisms by actor types, coordination structures (e.g., peer-to-peer, centralized), and strategies like role-based planning [8].

B. OPC AI framework

In traditional health care practices, a medical doctor often struggles with multitasking, working on every detail of daily operations and administrative work. Our OPC system in the proposal will help reduce their workload by introducing dynamic AI agents as assistants, which work autonomously to

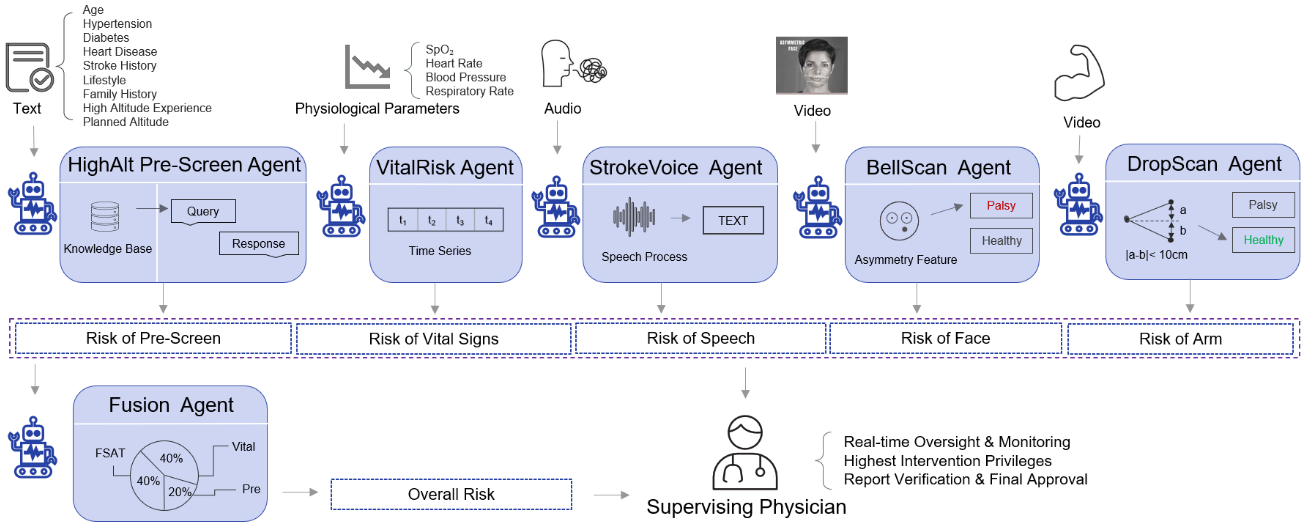

Fig. 1. Overview of the proposed OPC AI multi-agent system for health assistance.

perform specific roles and tasks, such as asking for medical history or measuring blood oxygen.

The proposed OPC AI workflow follows a “Person manager + Multi-Agent Collaboration” pattern: A person (manager, in our case, a doctor) as Core: The person is responsible for the high-level vision, strategy, creativity, and final decision-making, for example, to diagnose if a tourist will have a risk in developing a plateau stroke. Their unique human judgment, medical expertise, and critical thinking become the primary “leverage point” for health care.

C. Multi-modal fusion

Multi-modal fusion integrates information from multiple sources such as text, image, audio, and sensor data. By leveraging the complementary nature of different modalities, fusion techniques enable a more holistic understanding of complex problems [9]. While established approaches include early(data-level) and intermediate fusion(e.g., concatenation), the proposed system adopts late fusion (decision-level fusion) to maintain the functional independence of its six specialized agents.

D. Stroke assessment

The occurrence of stroke (a “brain attack”) in populations living or visiting high-altitude regions. It’s not a different disease, but the same condition with a distinct profile due to the unique physiological and environmental challenges of high altitude. Often, diagnosis involves: physical examination to assess symptoms and medical history, often using the FAST (Face, Arm, Speech, Time) test to identify common signs, and/or a CT scan and some blood flow tests. Research reveals that stroke in high-altitude regions, for example, Tibet, has a different profile compared to low-altitude areas [10].

III. Design of our proposed OPC AI system

To address the challenges of multi-modal data processing and the demand for high clinical response efficiency in high-altitude stroke risk screening, we propose a novel collaborative intelligent screening platform based on the OPC AI concept, which includes a supervising physician and six agents, shown in Fig. 1.

The system employs a layered architecture to establish a comprehensive decision support framework, which extends from underlying algorithmic support to high-level clinical implementation. As illustrated in Fig. 2, the overall architecture comprises the Model and Algorithm Support Layer, the Multi-Agent Collaborative Execution Layer, and the Application Layer.

A. Model and Algorithm Support Layer

As shown in the Model & Algorithm Support Layer in Fig. 2, a multi-modal infrastructure is designed, where LLMs serve as the logic hub for query semantic understanding and risk explanation generation. Specialized components perform feature extraction across three domains: the computer vision models for facial asymmetry and upper limb motion analysis, the speech models for voice transcription, and the signal processing models for time series analysis of vital signs.

B. Multi-Agent Collaborative Execution Layer

This layer consists of specialized agents that process data across different dimensions and output structured results to the Fusion Agent. To ensure modularity and seamless integration, each agent follows a single-responsibility design, functioning as a flexible node that independently outputs structured results to the collective network. This architecture supports the following specialized components:

*1) **HighAlt Pre-Screen Agent**:* The High-Altitude Pre-Screening Agent utilizes Baidu Qianfan’s autonomous planning agent for high-level semantic parsing based on user input, yet its operational logic is governed by a progress tracking table. To ensure clinical compliance, it decouples the agent’s semantic understanding from its decision-making,

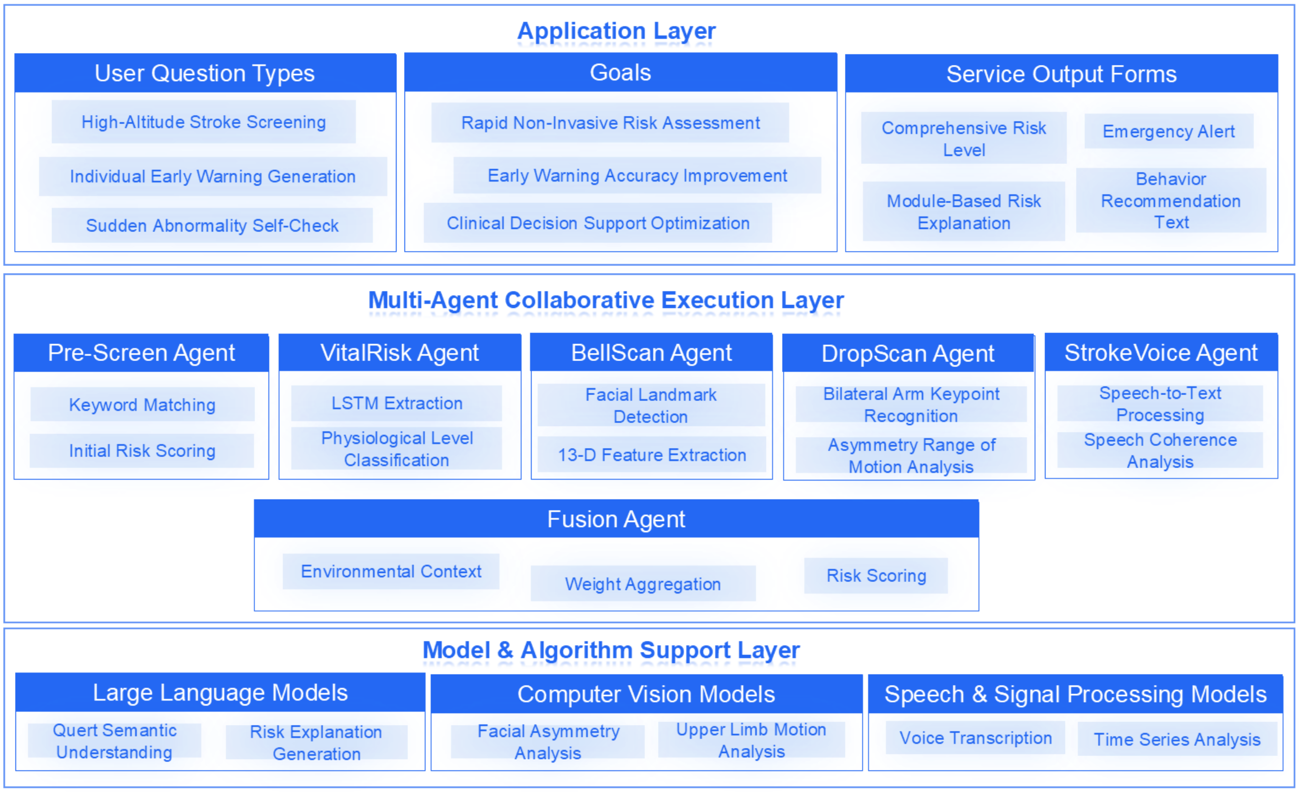

Fig. 2. Multi-Agent Health Assistance Architecture (OPC-Based)

while the LLM interprets unstructured user inputs, the state-aware table strictly confines the diagnostic flow to 9 predefined indicators. Every session is anchored to a predefined clinical query base (a fixed repository of standardized questions), preventing the model from generating arbitrary inquiries. The workflow follows a deterministic state-update logic: the agent must successfully map user responses to a specific metric in the table before the planning module triggers the next inquiry from the query base. This architecture ensures clinical rigor, facilitates breakpoint resumption, and maintains the factual grounding required for stroke pre-screening in high-altitude environments.

*2) **VitalRisk Agent**:* To handle physiological telemetry, we developed the VitalRisk Agent based on the PaddlePaddle framework for real-time risk assessment. This agent employs an LSTM network to capture the complex temporal dependencies of four vital indicators: blood oxygen saturation (SpO_2), heart rate (HR), blood pressure (BP), and respiratory rate (RR). By integrating a Multi-Head Attention mechanism, we further enhance the temporal feature representation for more accurate risk classification. To ensure seamless deployment, the agent is served via FastAPI RESTful APIs, providing a lightweight and high-speed response that outputs risk levels and confidence distributions in the structured JSON format, as detailed in Algorithm 1.

*3) **BellScan Agent**:* Specifically addressing the "F" (Facial asymmetry) in the FAST framework, the BellScan Agent

Algorithm 1 Physiological Risk Prediction with LSTM-Attention

Initialize: LSTM parameters, Multi-Head Attention weights, LayerNorm, and Classifier.
Define: Optimizer Adam and Loss Function $\mathcal{L}$ = CrossEntropy$(\hat{y}, y)$.

```
for epoch = 1 to E do
    Input: Vital signs sequence X = {spo2, hr, bp, rr}.
    H ← LSTM(X)    //Extract temporal features
    α ← Softmax(W_q Q · (W_k K)^T / √d_k)    //Attention weights calculation
    A ← Σ α_t H_t    //Context vector via weighted sum
    Z ← LayerNorm(H + A)    //Residual connection and normalization
    ŷ ← Classifier(Z_last)    //Risk score generation
    Compute loss L and update parameters via Backpropagation.
end for
return Trained model
procedure PREDICT(X_new)
    risk_scores ← Model(X_new)
    P ← Softmax(risk_scores)    //Probability distribution
    Level ← arg max(P)    //Map to Low, Medium, High
    return {Level, P}
end procedure
```

automates facial paralysis detection using a modular four-layer pipeline. After uploading video data via FastAPI, dlib is employed to extract 68 facial landmarks. Crucially, based on our preliminary video observations of stroke-related symptoms, we prioritized the mouth and eye regions as the most diagnostic areas for subtle asymmetry. To quantify these impairments, the system analyzes 30-frame sequences to compute 13 kinematic features, including range of motion and bilateral correlation. These metrics are then classified by a Support Vector Machine (SVM). To handle real-world tracking failures, we incorporated segmented sampling and zero-padding, ensuring the model remains robust against imperfect or intermittent video inputs. The first part of Algorithm 2 is the pseudoscope of this agent.

*4) **DropScan Agent**:* To address the "A" (Arm drift) component within the FAST framework, we implemented the DropScan Agent to automate the standard 10-second elevation test (Algorithm 2). Rather than relying on qualitative observation, the agent computes kinematic asymmetries by tracking the spatial coordinates of key upper-limb joints. We introduce a bi-shoulder reference baseline for data normalization, a step that allows the system to capture subtle downward trajectories often missed by the human eye. The detection logic is strictly aligned with clinical requirements: it triggers an automated stroke alert whenever the height discrepancy exceeds 10 cm or the stable elevation period falls below 9.5 s. Each generated report provides not only the risk level but also specific lateralization data and immediate clinical guidance.

*5) **StrokeVoice Agent**:* Designed for the "S" (Speech impairment) component of the FAST framework, the StrokeVoice Agent performs a multi-dimensional analysis of verbal interactions to identify early signs of dysphasia or dysarthria (Algorithm 2). Following a guided voice recording session, the agent initiates a dual-stream evaluation: it leverages the Baidu Qianfan for speech-to-text transcription while simultaneously monitoring real-time dysfluencies, such as pauses exceeding 2 seconds, filler words, and slurring. By synthesizing these transcription results with behavioral metrics, the agent calculates clinical scores across four key dimensions—word-finding difficulty, grammatical errors, clarity, and fluency. The process concludes with the generation of a report, applying predefined decision rules to determine the overall speech risk level.

*6) **Fusion Agent**:* The Fusion Agent functions as the decision-making core by synthesizing findings from 5 agents mentioned above. It implements a weighted fusion logic where each agent is assigned a baseline weight according to its diagnostic importance. The alert mechanism is triggered if all three FAST criteria are met. Additionally, an emergency alert is also issued if the case is classified as "High Risk" by the Pre-Screen Agent and at least one individual FAST criterion is satisfied. The process concludes by generating a structured report that summarizes the calculated risk level and provides targeted clinical advice for the user.

Algorithm 2 Face, Arm, Speech Assessment for Stroke Detection (FAST)

Require: Patient interaction data (visual observation V, response to commands A)
Ensure: Individual assessment results for Face, Arm, and Speech

```
// F - Face Assessment
procedure AssessFace(V)
    Ask patient to smile or show teeth to observe facial movement.
    Extract 68 landmarks; calculate MouthDiff, Corr, SpeedDiff
    if SVM_Predict(X_face) = Affected then
        FaceResult ← {AffectedRisk, Details}
    else
        FaceResult ← {UnaffectedRisk, Details}
    end if
end procedure
// A - Arm Assessment
procedure AssessArm(V)
    Ask patient to close eyes and hold both arms out, palms up, for 10 seconds.
    Track H_diff and T_hold via pose estimation.
    if H_diff > Thres and T_hold < 10s then
        ArmResult ← {AffectedRisk, Details}
    else
        ArmResult ← {UnaffectedRisk, Details}
    end if
end procedure
// S - Speech Assessment
procedure AssessSpeech(A)
    Ask patient to repeat a simple phrase (e.g., "The weather is fine and I want an apple.").
    Text ← ASR(A); Monitor T_p (pauses > 2s) and N_f (fillers).
    if Fluency ≤ 3 or WordFinding = True then
        SpeechResult ← {AffectedRisk, Details}
    else
        SpeechResult ← {UnaffectedRisk, Details}
    end if
end procedure
```

C. Application Layer

At the Application Layer, a supervising physician maintains central authority via a human-led and AI-agent-executed workflow, intervening in real-time for high-risk alerts. To ensure transparency, the system leverages its structured JSON outputs to synchronize information with subjects: once authorized by the physician, patients can access their specific risk indicators and diagnostic data. By distilling complex metrics into quantified levels and modular explanations, the system provides interpretable intelligence that assists both clinicians and users without replacing human judgment.

TABLE I
TESTING SCENARIOS

TextBook	Pre-Screen	VitalRisk	BellScan	DropScan	StrokeVoice	Fusion (Our System)
Normal	Medium	P(Low)=0.9916	False	False	Low	Low
Emergency	Low	P(High)=0.9954	True	True	Medium	High
Emergency	High	P(Medium)=0.9975	True	False	Medium	High
Emergency	Low	P(Low)=0.9919	True	True	High	Medium

IV. TESTING SCENARIOS FOR EVALUATION

The OPC AI agent system is tested. Table I lists four testing scenarios, and their risk levels for each agent. For example, the second scenario: a 20-year-old sedentary male smoker during his first acute exposure to 2,000m altitude. Prior to the plateau entry, the system initiates a proactive consultation request via the HighAlt Pre-Screen Agent. By conducting a table-driven dialogue (9 indicators), the agent obtains essential subject information to generate an initial "Low" risk JSON profile. Based on these identified lifestyle risks (e.g., smoking), the physician then presets customized monitoring frequencies for the subsequent ascent. At altitude, wearable data triggers the VitalRisk Agent for high-frequency monitoring (Period $T = 20$ min). It analyzes physiological parameters, outputting risk probability distribution $P(\text{risk})$. Upon acute symptoms, the physician activates the FAST protocol: BellScan, DropScan, and StrokeVoice agents run concurrently. The Fusion Agent weights and fuses their outputs (Fig. 3), delivering a final assessment to guide emergency intervention or hospital transfer.

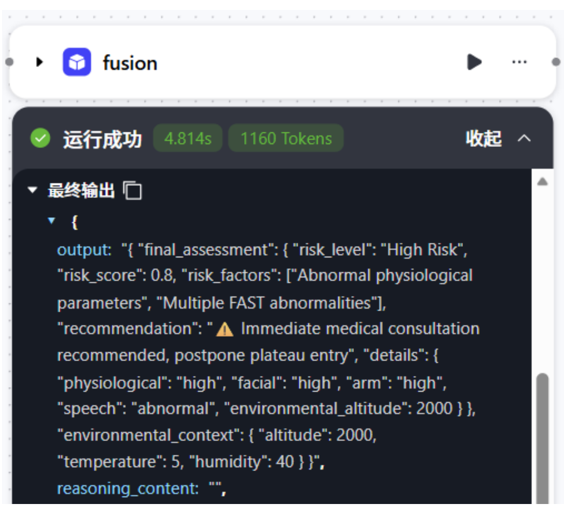

Fig. 3. Output of fusion agent for the second scenario

The system breaks down information silos, providing physicians with a continuous global view from early screening to emergency care. By pairing edge-based monitoring with the FAST protocol, it significantly cuts down the critical onset-to-decision time compared to traditional methods. Moreover, the autonomy of our agents allows a single doctor to manage multiple high-risk cases simultaneously—a vital capability for regions where medical resources are scarce.

V. CONCLUSION

This paper presents a multi-agent system for the high-altitude stroke screening via the OPC AI framework. By integrating six specialized agents (from structured pre-screening to FAST diagnostics), the system effectively assists physicians and users in evaluating the risk of stroke. The preliminary feasibility evaluation shows that this architecture enables remote monitoring of multiple users' health information simultaneously, while providing each user with a "digital physician" of their physician that monitors their health status on demand. Future work will shift from simulated data to clinical pilot studies.

REFERENCES

[1] C. Sha and L. Jun, "One-person companies gain traction in china with ai tools," People's Daily Online, Feb 2026. [Online]. Available: https://en.people.cn/n3/2026/0205/c90000-20422771.html

[2] Z. Kaiwei and L. Jun, "Ai allows china's digital natives to punch above their weight in tech scene," People's Daily Online, Jan 2026. [Online]. Available: https://en.people.cn/n3/2026/0129/c90000-20420273.html

[3] SIP, "Sip strives to cultivate an opc ecosystem to support solopreneurs," Suzhou Industrial Park Administrative Committee, Nov 2025. [Online]. Available: https://www.sipac.gov.cn

[4] M. G. S. V. Petridis, P., "Holistic ai in medicine; improved performance and explainability." *npj Digit. Med.*, vol. 9, no. 120, 2026.

[5] M. Y. Z. C. Soenksen, L.R., "Integrated multimodal artificial intelligence framework for healthcare applications," *npj Digit. Med.*, vol. 5, no. 149, 2022.

[6] H. Cui, Y. Du, Q. Yang, Y. Shao, and S. C. Liew, "Llmind: Orchestrating ai and iot with llm for complex task execution," *IEEE Communications Magazine*, vol. 63, no. 4, pp. 214–220, 2025.

[7] S. Agashe, Y. Fan, A. Reyna, and X. E. Wang, "LLM-coordination: Evaluating and analyzing multi-agent coordination abilities in large language models," in *Findings of the Association for Computational Linguistics: NAACL 2025*, L. Chiruzzo, A. Ritter, and L. Wang, Eds. Albuquerque, New Mexico: Association for Computational Linguistics, Apr. 2025, pp. 8053–8072. [Online]. Available: https://aclanthology.org/2025.findings-naacl.448/

[8] S. H. Shaikh, "Llm-based multi-agent systems: Frameworks, evaluation, open challenges, and research frontiers," in *Computational Intelligence*, F. Marcelloni, K. Madani, N. van Stein, and J. Filipe, Eds. Cham: Springer Nature Switzerland, 2026, pp. 149–170.

[9] F. Zhao, C. Zhang, and B. Geng, "Deep multimodal data fusion," *ACM Comput. Surv.*, vol. 56, no. 9, Apr. 2024. [Online]. Available: https://doi.org/10.1145/3649447

[10] Y. Lu, C. Zhuoga, H. Jin, X. Zhang, S. Zhou, W. Sun, N. Wang, Z. Zhou, N. Wang, and X. Huang, "Characteristics of acute ischemic stroke in hospitalized patients in tibet: a retrospective comparative study," *BMC Neurology*, vol. 20, no. 1, p. 380, Oct 2020.